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The Rising Entropy of English in the Attention Economy

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Abstract

We present evidence that the word entropy of American English has been rising steadily since around 1900, contrary to predictions from existing sociolinguistic theories. We also find differences in word entropy between media categories, with short-form media such as news and magazines having higher entropy than long-form media, and social media feeds having higher entropy still. To explain these results we develop an ecological model of the attention economy that combines ideas from Zipf's law and information foraging. In this model, media consumers maximize information utility rate taking into account the costs of information search, while media producers adapt to technologies that reduce search costs, driving them to generate higher entropy content in increasingly shorter formats.

Keywords: communication | language | entropy | attention economy | information foraging | Zipf's law

Introduction

Word entropy is a measure of the amount of repetition (low entropy) or novelty (high entropy) in word distributions. Empirical word distributions typically follow Zipf's law, which describes a power law between a word's observed frequency and that word's rank in the frequency distribution [1]. This empirical power law is remarkably stable with an exponent around 1 [2, 3, 4]. The stability of Zipf's law suggests some underlying mechanism, and Zipf himself hypothesised a *principle of least effort* between speakers and listeners. More recently this principle has been expanded to show that power laws in word distributions can emerge from a balance between maximising the benefits of receiving highly informative messages (preferred by listeners) and minimising the costs of generating high word entropy text (preferred by speakers) [4].

In recent times this balance between the efforts of listeners and speakers has changed. Modern communication systems have transformed the way that we share and consume information, in particular by increasing the accessibility of information [5]. In the words of Herbert Simon this creates a "poverty of attention" [6], such that media producers must compete for the limited resource of human attention [7, 8, 9]. This dynamic has been called *the attention economy*, a combination of forces influencing the production and consumption of information, with consequences including a shortening collective attention span [10]. If information adapts to the balance between the preferences of media producers and consumers, then increased competition for attention tips the balance toward the preferences of the consumers. That is, information markets (the distribution of available content) should rise in information density, and specifically, entropy.

We can envision this adaptive process in terms of *information foraging* [11, 12]. Information foraging describes how people search for and consume information in different environments, including web browsing [13] software debugging [14, 15, 16], and the design of information and social environments [13, 16, 17]. The basic rationale of this approach is borrowed from ecological models of foraging, which have been shown to be appropriate to a wide range of search problems ranging from spatial foraging to

cultural evolution [18]. Indeed, handling the exploration versus exploitation trade-off that is common to all of these environments has been proposed to be a defining selective force in the evolution of cognition [19, 20].

In what follows, we first investigate the evolution of information across a wide variety of media sources over the last two centuries, a time marked by increasing media competition. We show how this reveals a characteristic pattern of rising entropy that affects different categories of media in different ways (e.g., books versus news versus social media). We then create a model of the attention economy that expands on existing models of information foraging to incorporate competition for human attention between media producers. This model explains both the general increase in word entropy and the differences in word entropy across categories.

Materials and Methods

Text Corpora

To investigate the recent history of information evolution we examine a variety of text corpora. The Corpus of Historical American English (COHA) [21] has $116,614$ texts spanning the 1810s to 2000s, balanced between categories of fiction ($n = 11,010$), non-fiction ($n = 2,635$), news ($n = 41,677$) and magazines ($n = 61,292$). The Corpus of Contemporary American English (COCA) has over 150,000 texts from between 1990 to 2008 split between fiction, popular magazines, newspapers, academic journals and spoken word [22]. For our analysis we used a publicly available sample of COCA with $2,362$ texts split between categories of fiction ($n = 275$), academic journals ($n = 266$), news ($n = 872$) and magazines ($n = 949$). The British National Corpus (BNC) contains $8,098$ texts from between 1960 and 1993 including written categories of fiction ($n = 904$), academic prose ($n = 994$), newspapers ($n = 972$), non-academic prose and biography, other published materials and unpublished materials [23]. Fiction and newspapers are common categories across the corpora. Magazines are a common category between COHA and COCA. We grouped as non-fiction the categories of COHA non-fiction, COCA academic journals and BNC academic prose.

The text sample data was cleaned before analysis in a standard way [24]. COHA and COCA are similar formats and so followed the same procedure. For both:

- Stripped any headers not a part of the main text samples.
- Removed any XML text tags.
- Removed any sentences that contained "@" symbols. COHA and COCA randomly replace words with @ symbol in groups of ten for copyright reasons [25].
- Removed apostrophes and extra whitespace.
- Used python's natural language toolkit (nltk) package to convert text to tokens [26].
- Selected the last 2000 tokens (words) of the text sample for processing. This avoids, as much as possible, anomalous text that sometimes appears at the start of text samples such as a contents section.

For the BNC data, python's natural language toolkit package comes with a BNC corpus reader [26], which was used to extract tokens. The only other treatment was to remove extra whitespace and apostrophes as with COCA and COHA.

The cleaned datasets had the following surviving sample counts with $N \geq 2000$ words:

- COHA total $n = 22,253$. Fiction $n = 8,164$, non-fiction $n = 2,046$, news $n = 725$, magazines $n = 11,318$.
- COCA total $n = 985$. Fiction $n = 167$, non-fiction $n = 166$, news $n = 39$, magazines $n = 133$.
- BNC total $n = 1,319$. Fiction $n = 447$, non-fiction $n = 477$, news $n = 395$.

The COHA dataset was analysed as a timeseries, so requires a large number of samples. The BNC and COCA, being corpora from much narrower time ranges, were analysed as distributions and as such require less samples.

Social Media Data

We also investigated social media. The Twitter dataset consisted of 1.6 million tweets scraped from the twitter API between April and June 2009 [27] and available online at `https://www.kaggle.com/kazanova/sentiment140`. To simulate a Twitter feed the tweets were chronologically collated to create $n = 1000$ text samples with $N \geq 2000$ words each.

For Reddit, we aimed to capture text samples that were representative of the text a user would see when visiting the site. To achieve this we used Reddit's API to download posts from the Reddit homepage feed at `https://oauth.reddit.com/.json`. Following Reddit's API rules, we first registered an app and all requests were authenticated with OAuth2. We downloaded 10,000 posts in JSON format in this way. We extracted the text from the posts and combined them to create $n = 90$ text samples with length $N \geq 2000$ words each. During processing we found a small number of non-English posts in the feed, which were removed.

The social media data was then cleaned:

- Removed apostrophes and extra whitespace.
- Removed any urls.
- Removed hashtags and usernames i.e. any words containing "@" or "#".
- Used python's natural language toolkit (nltk) package [26] to convert the collated samples into a list of tokens, and the last 2000 tokens taken.

Social media statuses are by nature short and are usually much smaller than $N = 2000$ words, and lexical measures of short text samples have little meaning. Our analysis is on the level of the social media feed and we generated large text samples through the collation of posts. This kind of collation will naturally create text samples with high lexical diversity. This isn't a flawed analysis — the high information density of a social media feed is related to the collation of statuses and how people actually consume social media.

Measures of information evolution

Information evolution is measured using unigram word entropy. For robustness we also analysed the type token ratio and Zipf exponent of text samples, which are also measures of lexical diversity [2]. The lexical measures are all sensitive to sample size, so we used truncated text samples to $N = 2000$ words.

Empirical unigram word entropy, H_1, is a function of the relative frequencies of each word, f_i, summed over the set of W unique words in the text sample. We use the maximum likelihood or plug-in estimator, which has the benefit of being simple and well known. And it has been shown to correlate well with more advanced estimators [28].

$$H_1 = -\sum_{i=1}^{W} f_i log_2 f_i \,. \tag{1}$$

Type token ratio (TTR) is the number of unique words (types) divided by the total words (tokens) in a text sample.

$$TTR = \frac{\#types}{\#tokens} \,. \tag{2}$$

Words in natural language are typically approximately distributed as a power law distribution between type frequency, f_i, and type rank in that frequency distribution, $r(f_i)$ [29]. This power law is parameterised by the Zipf exponent, α, which describes the steepness of the distribution in log space. Maximum likelihood estimation was used to estimate the Zipf exponent [29]. This estimator has the benefit of being widely used and well known. It shows bias (as do all Zipf estimators [30]), but the bias is systematic so can be ignored for the purpose of comparision of text samples.

$$f_i \propto r(f_i)^{-\alpha} \,. \tag{3}$$

Each of the measures were applied once to the same set of distinct text samples.

Timeseries Breakpoint Analysis

The Corpus of Historical American English (COHA) provides historical text samples across fiction, non-fiction, news and magazines categories. The type token ratio, word entropy and Zipf exponent were calculated for each text sample with over 2000 words.